

ABHIJIT DESHMANE

Software Engineer · Full-Stack Developer · Systems & AI Builder

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PROFESSIONAL SUMMARY

Full-stack software engineer and Computer Science undergraduate experienced in architecting production-grade web platforms, autonomous browser agents, and developer tooling. Proven track record designing resilient distributed systems, including DAG-based workflow execution engines, multiplayer CRDT canvases, and terminal-native AI applications using Next.js, Bun, TypeScript, and PostgreSQL. Solved 160+ algorithmic problems on LeetCode.

SKILLS

Programming Languages: C++, TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

Frontend Engineering: React 19, Next.js (App Router), React Native, Tailwind CSS, React Flow, Responsive Design

Backend & Runtimes: Node.js, Bun, Express.js, Hono, RESTful APIs, Server-Sent Events (SSE), Webhooks

Databases & DevOps: PostgreSQL (Neon Serverless), MongoDB, Prisma ORM, Drizzle ORM, Docker, Linux, Git, GitHub Actions

Systems & Architecture: Directed Acyclic Graphs (DAG), Inngest Durable Tasks, Liveblocks (CRDT), Trigger.dev, AES-256

Core Fundamentals: Data Structures & Algorithms (DSA), OOP (C++), DBMS, Operating Systems (OS), Computer Networks

PROJECTS

Orchify | [GitHub](#)

Cloud-Native Visual Workflow Automation Platform

- **Engineered a distributed DAG execution engine** with topological sorting and cycle detection, guaranteeing deterministic, sequential step execution for multi-node visual workflows.
- **Implemented zero-loss durable execution** via Inngest step functions, handling automated retries, rate limits, and asynchronous webhooks across Stripe, Google Forms, Slack, and Discord.
- **Streamed real-time node state telemetry** (`loading`, `success`, `error`) over SSE via Inngest to a React Flow canvas with sub-100ms latency, secured with AES-256-GCM encrypted API key storage.

Technologies / Tools Used: Next.js 15, React 19, TypeScript, PostgreSQL, Prisma, React Flow, Inngest, tRPC v11, Polar.sh

Navix | [GitHub](#)

Collaborative AI Browser Automation & Web Agent Platform

- **Architected an autonomous browser agent engine** using Stagehand and Browserbase sandboxes to replace brittle CSS selectors with natural language semantic web actions driven by Gemini Flash.
- **Engineered a multiplayer visual canvas** using Liveblocks CRDTs and React Flow, enabling concurrent multi-user graph editing, live cursor presence, and conflict-free state synchronization.
- **Orchestrated resilient background pipelines** via Trigger.dev v4 durable tasks for dynamic variable interpolation (`{{node.path}}`), streaming `.m3u8` HLS replays via `hls.js`.

Technologies / Tools Used: Next.js 16, React 19, TypeScript, PostgreSQL, Drizzle ORM, Browserbase, Stagehand, Liveblocks, Trigger.dev v4

AgentForge | [GitHub](#)

Terminal-First Autonomous AI Coding Assistant

- **Developed a 60 FPS terminal-native coding assistant** using OpenTUI and React 19, delivering zero-latency keyboard navigation, animated thinking blocks, and token streaming in the CLI.
- **Architected a Dual-Mode execution engine** featuring a safe read-only **PLAN** mode for AST exploration and an autonomous **BUILD** mode with strict path containment (`resolveInsideCwd`).
- **Engineered a decoupled client-server architecture** via Bun + Hono HTTP & SSE gateway for multi-model routing (Claude with extended thinking & OpenAI) with credit accounting via Polar.sh.

Technologies / Tools Used: Bun, TypeScript, React 19, OpenTUI, Hono, Prisma, PostgreSQL, Vercel AI SDK

PROBLEM SOLVING & ACHIEVEMENTS

- **LeetCode Problem Solving:** Solved **160+ Data Structures & Algorithms problems** ([Profile](#)) with consistent focus on **Graph Theory, Trees, Recursion, and Dynamic Programming**.
- **Algorithmic Optimization:** Proficient in computational complexity, regularly writing solutions optimized for $O(N \log N)$ or $O(N)$ runtime performance and minimal auxiliary space.
- **Systems Track Record:** Architected 3 end-to-end production systems covering distributed DAG execution, CRDT collaborative canvases, and low-latency terminal TUI rendering.

EDUCATION

Dr. Babasaheb Ambedkar Technological University (BATU)

Bachelor of Technology in Computer Science and Engineering

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (C++), DBMS, Operating Systems, Computer Networks, System Design Basics.

Aug 2024 – Jun 2028

CGPA: 8.41 / 10.0